

Latent Geometry of Time Series Foundation Models: Structure, Stability, and Reliability

Abstract

Time series foundation models (TSFMs) achieve strong zero-shot transfer across diverse domains, yet the structure of their learned representations remains uncharacterised. We introduce the **Latent Geometry Framework (LGF)**, a diagnostic toolkit that maps TSFM representation spaces through three complementary lenses, operating entirely on frozen pretrained weights without retraining or label access. First, we estimate intrinsic dimensionality (ID) layer by layer via the TwoNN estimator, revealing how effective dimension evolves through the network. Second, we quantify manifold regularity through three perturbation-grounded metrics—Latent Cohesion Score (LCS), Temporal Stability Index (TSI), and Generalised Stability under Perturbation (GSP)—computed against STL-derived semantic trend labels. Third, we measure a geodesic reliability protocol correlating manifold proximity with forecast error, yielding a label-free deployment-time trustworthiness signal. Applying LGF to TimesFM 2.0 Das et al. [2024] and Chronos-T5-Large Ansari et al. [2024] across four standard benchmarks (Electricity, ETTh1, Weather, METR-LA), we find that TSFMs compress representations to significantly lower intrinsic dimension than supervised baselines and randomly-initialised controls (compression ratio **X.XX** vs. **X.XX**), organise embeddings into semantically coherent clusters (LCS = **X.XX** vs. **X.XX**, $p < 0.001$), and exhibit geodesic-forecast correlation $\rho_s = \mathbf{X.XX}$ enabling zero-shot reliability estimation. Linear probes recover temporal primitives from frozen TSFM embeddings with **X.XX%** balanced accuracy versus **X.XX%** for supervised baselines. LGF provides the first geometric account of what TSFMs learn and a principled criterion for assessing zero-shot forecast trustworthiness.

1. Introduction

Foundation models for time series—TimesFM Das et al. [2024], Chronos Ansari et al. [2024], MOMENT Goswami et al. [2024], and others Woo et al. [2024], Rasul et al. [2024]—are trained on large heterogeneous corpora spanning energy, climate, finance, and traffic. Their zero-shot transfer capability is broadly documented, yet the internal structure of their learned representations is entirely uncharacterised. We do not know whether TSFM embedding spaces are geometrically regular, whether they encode interpretable temporal concepts, or whether their latent geometry carries any signal about forecast reliability.

This gap is not purely academic. Zeng et al. [2023] demonstrated that a one-layer linear model matches or outperforms deep Transformer forecasters on five standard long-horizon benchmarks, establishing that architectural complexity alone does not explain forecasting quality. If complexity is insufficient, characterising *what* large pretrained models actually encode—rather than simply measuring whether they outperform baselines on held-out data—becomes essential for understanding and trusting their outputs. Geometric analysis of representations offers precisely this characterisation.

In vision, Ansuini et al. [2019] showed that image classifiers compress representations to low-dimensional manifolds in terminal layers, with the compression ratio tracking generalisation quality. In language, Reif et al. [2019] demonstrated that BERT organises contextual embeddings into geometrically separable syntactic and semantic subspaces. Raghu et al. [2017] found that network layers stabilise at different training stages, reshaping understanding of what depth achieves. No equivalent geometric characterisation exists for time series models.

In this work, we introduce the **Latent Geometry Framework (LGF)**, a reusable suite for characterising TSFM representation spaces via frozen inference. LGF combines layer-wise intrinsic dimensionality profiling, three perturbation-grounded manifold regularity metrics with semantically grounded labels, and a geodesic protocol that links manifold proximity to forecast error. Our main contributions are summarised as follows:

- We propose **LGF**, the first geometric analysis toolkit for TSFM representation spaces, combining TwoNN intrinsic dimensionality profiling, three perturbation-grounded metrics (LCS, TSI, GSP) with STL-derived semantic labels, and a geodesic-forecast reliability protocol—all operating on frozen weights without retraining.
- We present the **first systematic empirical characterisation** of TSFM latent geometry across two architectures and four datasets, showing that TSFMs develop significantly lower-dimensional and more semantically coherent representations than supervised baselines, and that this structure is absent in randomly-initialised controls, demonstrating it is learned rather than architectural.
- We show that **frozen TSFM embeddings linearly encode temporal primitives**—trend direction, volatility, and seasonality—localised to specific layer ranges, and that

geodesic proximity in the embedding space predicts forecast error, enabling label-free reliability estimation at deployment time.

We evaluate across two TSFM architectures (decoder-only: TimesFM 2.0; encoder-decoder: Chronos-T5-Large) and four benchmark datasets with three supervised baselines and a random-weight control. Findings are observational; we characterise geometric structure without claiming it causes forecasting quality.

2. Background and Related Work

2.1. Time Series Foundation Models

The past two years have produced a distinct class of models trained on large, heterogeneous collections of time series with the aim of zero-shot or few-shot transfer across domains. These models vary along two architectural axes: tokenisation strategy and sequence-to-sequence objective.

Patch-based encoder-decoder models. MOMENT Goswami et al. [2024] applies masked patch modelling to sub-series segments, learning a representation that can be adapted to forecasting, classification, and anomaly detection. Moirai Woo et al. [2024] extends this with a multi-patch-size architecture that trains a single model to handle variable frequencies.

Decoder-only models. TimesFM Das et al. [2024] treats forecasting as next-patch prediction, training a 500M-parameter decoder-only model on a corpus spanning Google Trends, Wikipedia pageviews, and synthetic series. Timer Liu et al. [2024] follows a similar autoregressive design at smaller scale.

Language-model-inspired models. Chronos Ansari et al. [2024] quantises time series values into discrete tokens and fine-tunes T5 Raffel et al. [2020] encoder-decoder architectures on a curated public corpus, treating forecasting as a sequence-to-sequence language task. Lag-Llama Rasul et al. [2024] adapts LLaMA with lag features as covariates.

Comprehensive surveys of this space are provided by Jin et al. [2023] and Liang et al. [2024]. Despite broad empirical coverage, neither survey addresses the internal geometric structure of learned representations—the question this paper targets.

The case against complexity. Before attributing zero-shot transfer to any internal property of TSFMs, a direct challenge must be addressed. Zeng et al. [2023] demonstrated that a simple one-layer linear model matches or outperforms Transformer-based forecasters on five standard long-horizon benchmarks, including ETT and Weather. This result does not directly refute TSFMs—linear models were not tested in zero-shot settings, and the comparison is against models trained per-dataset—but it establishes that Transformer complexity is neither necessary nor sufficient for forecasting quality in supervised settings. Our geometric analysis is designed with this sceptical position in mind: rather than asserting TSFMs are better, we ask whether

their latent representations are geometrically distinct from supervised baselines trained on the same data, and whether any such distinction correlates with forecast reliability.

2.2. Geometric Analysis of Neural Representations

The internal geometry of neural network representations has been studied systematically in vision and language models, establishing a set of analytical tools directly applicable to our setting.

Intrinsic dimensionality. Ansuini et al. [2019] showed that deep image classifiers exhibit a characteristic ID profile: representations expand to high effective dimensionality in early layers and compress toward a low-dimensional manifold in the final layers, with compression correlating with generalisation quality. Birdal et al. [2021] extended this to language models, finding that low intrinsic dimensionality of the fine-tuning landscape explains why large pretrained models are efficiently adapted with few parameters. We apply the same analysis to TSFMs, where the relevant question is whether the compression pattern reflects meaningful temporal abstraction or merely the inductive bias of the Transformer architecture.

Geometry of language model representations. Reif et al. [2019] applied geometric analysis to BERT, finding that contextual embeddings form geometrically separable clusters corresponding to syntactic and semantic categories. Kornblith et al. [2019] introduced Centred Kernel Alignment (CKA) to compare representation geometry across architectures, showing that wider and deeper networks converge to similar representations in intermediate layers. No equivalent geometric analysis exists for time series models.

Representational similarity. Raghu et al. [2017] introduced SVCCA to track representation geometry through training, demonstrating that lower layers stabilise early while upper layers continue evolving—a pattern relevant to our layer-wise ID profile analysis.

2.3. Explainability for Time Series

Existing XAI methods for time series fall into two categories: attribution methods that assign importance scores to input timesteps (LIME, SHAP, integrated gradients), and perturbation-based methods that measure model sensitivity to input modifications Ismail et al. [2020], Theissler et al. [2022]. Both categories operate at the input level and produce post-hoc explanations for individual predictions. Neither provides a structural account of what a model has learned to encode.

Our approach is fundamentally different: we analyse the *representation space* itself—its dimensionality, stability, and semantic organisation—rather than attributing individual predictions. This distinction aligns GEF with mechanistic analysis rather than attribution-based XAI, and the two approaches are complementary rather than competing.

2.4. Linear Probing and Temporal Primitives

Linear probing Alain and Bengio [2017] tests whether a target property is linearly decodable from frozen intermediate representations, providing a controlled measure of what information is encoded without confounds from end-to-end training. In NLP, probing has revealed that BERT encodes syntactic structure in intermediate layers and semantic structure in upper layers Reif et al. [2019]. We apply this methodology to test whether TSFM embeddings encode three interpretable temporal primitives: trend direction, volatility regime, and seasonality—properties that are trivially computable from the input but non-trivially encoded if the representation space organises them as separable geometric subspaces.

2.5. Theoretical Motivation

A classical result in dynamical systems theory offers an intuitive lens. Takens’ embedding theorem Takens [1981] states that delay embeddings of scalar observations from a smooth deterministic dynamical system reconstruct the system’s attractor manifold, up to diffeomorphism. In other words, finite windows of a time series generated by a dynamical system already lie on a low-dimensional manifold in delay-embedding space.

We invoke this result as *motivating intuition only*. The theorem’s conditions—smooth, noise-free, deterministic dynamics—are not satisfied by the heterogeneous real-world series (electricity consumption, traffic speed, meteorological observations) studied here. Our hypothesis, stated without claiming the theorem’s authority, is that TSFMs trained on diverse corpora have learned to approximate the manifold structure of temporal data in their latent space, and that this approximation is measurable via the geometric tools described in Section 3. Whether this constitutes genuine manifold learning or a looser form of distributional compression is precisely what our experiments test.

3. Latent Geometry Framework

3.1. Problem Formulation

Let $f_\theta : \mathbb{R}^L \rightarrow \mathbb{R}^D$ denote the mapping from an input context window $w \in \mathbb{R}^L$ to a sequence embedding $z = f_\theta(w) \in \mathbb{R}^D$ produced by a frozen pretrained TSFM. For a TSFM with $\mathcal{L}$ transformer layers, we additionally denote the intermediate mapping at layer ℓ as $f_\theta^\ell : \mathbb{R}^L \rightarrow \mathbb{R}^{T \times D_\ell}$, where $T = L/P$ is the number of patch tokens and D_ℓ is the hidden dimension at layer ℓ . The sequence embedding is obtained by mean-pooling over tokens: $z^\ell = \frac{1}{T} \sum_{t=1}^T f_\theta^\ell(w)_t$.

Given a dataset $\mathcal{W} = \{w_i\}_{i=1}^N$ of windows drawn from a target time series and the corresponding forecast errors $\mathcal{E} = \{e_i\}_{i=1}^N$ produced by the TSFM at a fixed horizon H , GEF addresses three questions:

1. **Structure:** Does the embedding set $\mathcal{Z} = \{z_i\}_{i=1}^N$ form a low-dimensional, geometrically regular manifold, and does this regularity differ from that of supervised baseline

models?

2. **Semantics:** Do geometric clusters in $\mathcal{Z}$ correspond to interpretable temporal properties of the input windows?
3. **Reliability:** Does proximity in $\mathcal{Z}$, measured along the manifold, predict forecast error magnitude?

3.2. Temporal Representation Manifold

Definition 1 (Temporal Representation Manifold). Let $\mathcal{P}$ be a probability distribution over time series windows of length L . The *Temporal Representation Manifold* (TRM) of a TSFM f_θ with respect to $\mathcal{P}$ is the image $\mathcal{M}_\theta = f_\theta(\text{supp}(\mathcal{P})) \subset \mathbb{R}^D$.

The TRM is not assumed to be a smooth manifold in the differential-geometric sense; Definition 1 is a set-theoretic statement about the image of the encoder. Whether $\mathcal{M}_\theta$ has low intrinsic dimensionality, smooth local geometry, and semantically organised cluster structure are *empirical questions* that GEF is designed to answer. In practice, $\mathcal{M}_\theta$ is approximated by the finite embedding set $\mathcal{Z}$ extracted from a sample $\mathcal{W} \sim \mathcal{P}$.

3.3. Geometric Regularity Metrics

We quantify three aspects of TRM regularity using perturbation-grounded metrics. These metrics apply established geometric machinery to a domain that has not previously been characterised through this lens; we do not claim the underlying methods (silhouette scoring, cosine stability) as contributions.

Semantic labels. A cohesion metric is only as meaningful as the labels it operates on. For each window w_i we derive a trend label $y_i \in \{\text{up}, \text{flat}, \text{down}\}$ from STL decomposition Cleveland et al. [1990]: extract the trend component, compute the OLS slope $\hat{\beta}_i$, and assign

$$y_i = \begin{cases} \text{up} & \hat{\beta}_i > \delta \cdot \sigma_{w_i} \\ \text{down} & \hat{\beta}_i < -\delta \cdot \sigma_{w_i} \\ \text{flat} & \text{otherwise,} \end{cases} \quad (1)$$

where $\sigma_{w_i} = \text{std}(w_i)$ and $\delta = 0.05$ (sensitivity to $\delta \in \{0.02, 0.05, 0.10\}$ is assessed in Section ??). Labels are derived entirely from window values, independently of any model.

Latent Cohesion Score (LCS). LCS measures whether embeddings from windows sharing the same trend label cluster together in $\mathbb{R}^D$:

$$\text{LCS}(\mathcal{Z}, \mathbf{y}) = \frac{1}{N} \sum_{i=1}^N \frac{b(i) - a(i)}{\max(a(i), b(i))}, \quad (2)$$

where $a(i)$ is the mean intra-cluster distance and $b(i)$ the mean nearest-cluster distance for z_i under labels $\mathbf{y}$. $\text{LCS} \in [-1, 1]$; values near +1 indicate tight, well-separated clusters.

Temporal Stability Index (TSI). For perturbation type $\mathcal{P}$, TSI measures how much the embedding of a window changes when its input is corrupted:

$$\text{TSI}_{\mathcal{P}}(\mathcal{Z}) = \frac{1}{N} \sum_{i=1}^N \frac{z_i \cdot \tilde{z}_i}{\|z_i\| \|\tilde{z}_i\|}, \quad \tilde{z}_i = f_{\theta}(\mathcal{P}(w_i)). \quad (3)$$

TSI $\in [-1, 1]$; values near +1 indicate a geometrically stable TRM under realistic input corruption. We evaluate three perturbation types: $\mathcal{P} \in \{\text{AddNoise}(\sigma=0.02), \text{Dropout}(p=0.05), \text{TimeWarp}(k=3)\}$.

Generalised Stability under Perturbation (GSP).

$$\text{GSP}_{\mathcal{P}} = \text{LCS}_{\text{clean}} - \text{LCS}_{\mathcal{P}}, \quad (4)$$

where $\text{LCS}_{\mathcal{P}}$ is computed on perturbed embeddings $\{\tilde{z}_i\}$. Low GSP (near zero) means the semantic cluster structure of the TRM is preserved after input perturbation.

3.4. Intrinsic Dimensionality Profile

The TwoNN estimator Facco et al. [2017] estimates the local intrinsic dimensionality of a point cloud from the ratio of first and second nearest-neighbour distances, without assuming a global parametric form:

$$\hat{d} = \frac{-N}{\sum_{i=1}^N \ln \mu_i}, \quad \mu_i = \frac{\|z_i - z_i^{(2)}\|_2}{\|z_i - z_i^{(1)}\|_2}, \quad (5)$$

where $z_i^{(1)}, z_i^{(2)}$ are the first and second nearest neighbours of z_i in Euclidean distance. We compute $\hat{d}^\ell$ at every layer $\ell \in \{0, \dots, \mathcal{L}\}$ to obtain the *ID profile*—the evolution of effective dimensionality through the network. The *compression ratio* $\hat{d}^0 / \hat{d}^\mathcal{L}$ summarises how much the terminal layer has reduced the effective dimension relative to the embedding layer.

3.5. Geodesic Reliability Protocol

Let $G = (V, E, w)$ be the k -nearest-neighbour graph on $\mathcal{Z}$ with edge weights equal to Euclidean distances. The geodesic distance between windows w_i and w_j is the shortest-path distance $d_G(i, j)$ on G (computed via Dijkstra’s algorithm), approximating the geodesic distance along $\mathcal{M}_\theta$.

For each window w_i , let $j_i^* = \arg \min_{j \neq i} d_G(i, j)$ be its nearest manifold neighbour, and let e_i be the TSFM’s per-window forecast MAE. We measure the Spearman rank correlation:

$$\rho_s = \text{Spearman}(d_G(i, j_i^*), |e_i - e_{j_i^*}|)_{i=1}^N. \quad (6)$$

A positive ρ_s means windows that are close on the TRM produce similar forecast errors—the TRM is geometrically consistent with forecast quality. This provides a **label-free reliability signal**: given a new window, its geodesic distance to the nearest

in-distribution window in $\mathcal{Z}$ indicates how much to trust the forecast.

We report ρ_s , its 95% confidence interval (Fisher’s z -transform), and the two-tailed p -value for the null hypothesis $\rho_s = 0$.

3.6. Theoretical Framing

The following proposition motivates why a low-dimensional, stable TRM should correlate with forecast consistency. It is stated as a conjecture grounded in the geometry of smooth functions and is tested empirically in Section ??.

Proposition 1 (Manifold Consistency, informal). *Let f_θ be a Lipschitz-continuous encoder with Lipschitz constant K . If two windows w_i, w_j satisfy $\|w_i - w_j\|_2 \leq \varepsilon$, then $\|z_i - z_j\|_2 \leq K\varepsilon$. If the forecast head g_θ is similarly Lipschitz with constant K' , then the expected absolute forecast difference satisfies $\mathbb{E}[|e_i - e_j|] \leq (K + K')\varepsilon + \eta$, where η accounts for aleatoric noise.*

Proposition 1 is not novel—it follows from elementary Lipschitz composition—but it makes the geometric claim precise: if the TRM is smooth (embeddings of similar windows are close), then similar windows receive similar forecasts, and forecast errors co-vary. The empirical question is whether the *manifold metric* d_G is a better predictor of forecast-error similarity than Euclidean distance in input space, which would indicate that the TRM encodes more than a trivial proximity structure.

3.7. LGF Summary

Figure 1 illustrates the LGF analysis pipeline. Given a frozen TSFM and a set of windows from a target dataset, GEF proceeds through three stages: (1) extraction of layer-wise hidden states and computation of the ID profile; (2) computation of LCS, TSI, and GSP under STL-derived semantic labels and three perturbation types; and (3) construction of the geodesic reliability protocol via the k -NN graph and Spearman correlation with forecast error. All stages require only frozen inference—no retraining or gradient access.

4. Experiments

4.1. Experimental Setup

Models. The current executable pipeline evaluates three supervised baselines trained from scratch: a two-layer **LSTM** (hidden 128), a three-block **TCN** (channels 128, kernel 3), and a three-layer **Transformer encoder** ($d=128$, 8 heads, pre-norm), all with AdamW ($\text{lr}=3 \times 10^{-4}$) and walk-forward cross-validation (3 folds).

Datasets. The loader supports four long-horizon benchmarks (Table 1): Electricity, ETTh1, Weather, and METR-LA.

Extraction protocol. For each fold, the script stores evaluation windows and uses their flattened vectors as proxy latent

[Figure: GEF three-stage pipeline diagram]
 Extraction $\rightarrow$ Regularity (LCS/TSI/GSP/ID) $\rightarrow$
 Reliability (ρ_s)

Figure 1. The Latent Geometry Framework (LGF) operates on frozen TSFM weights across three stages: latent extraction and ID profiling, geometric regularity measurement, and geodesic reliability correlation. No fine-tuning or label access is required at inference time.

Table 1. Benchmark datasets across four temporal domains.

Dataset	Domain	Length	Freq.
Electricity	Energy	26,304	Hourly
ETTh1	Temperature	17,420	Hourly
Weather	Meteorology	52,696	10-min
METR-LA	Traffic	34,272	5-min

representations for geometry diagnostics. Perturbation operators are then applied to these latent vectors (noise, masking, time warp) to compute stability-oriented metrics.

Statistical reporting. Current outputs are point estimates saved to CSV files. Bootstrap confidence intervals and significance testing are planned but not yet included in the default run script.

4.2. Qualitative Manifold Structure

We project latent vectors with UMAP (PCA fallback in implementation) and export: (i) 2D scatter figures colored by temporal index, and (ii) interactive 3D HTML plots with a chronological trajectory overlay. These figures are used as qualitative diagnostics of latent organisation for each baseline model.

4.3. Manifold Regularity

Table 2 reports the three implemented regularity metrics: LCS, TSI, and GSP. In the current code, LCS uses periodic labels derived from sample index ($i \bmod 8$), while TSI/GSP are computed on perturbed latent vectors.

These metrics are currently used as implementation checks and comparative descriptors across baseline branches. Because label construction is heuristic (phase bins), results should not yet be interpreted as semantic trend separation.

Table 2. Regularity metrics produced by the current baseline pipeline. Values shown are placeholders for rerun outputs.

Model	LCS	TSI	GSP
LSTM	X.XX	X.XX	X.XX
TCN	X.XX	X.XX	X.XX
Transformer	X.XX	X.XX	X.XX

4.4. Current Scope and Pending Modules

The current experiments cover baseline training, proxy-latent extraction, visualisation, and perturbation-based regularity metrics. The following experiment modules are not yet integrated in the default pipeline: TSFM hidden-state benchmarking, intrinsic dimensionality profiling, geodesic reliability analysis, temporal primitive probing, and failure-case characterisation. These modules are treated as planned extensions rather than reported empirical results in this version.

5. Discussion and Conclusion

We introduce LGF, the first geometric characterisation of TSFM representation spaces, revealing that these models organise temporal data into low-dimensional, semantically coherent manifolds that encode interpretable temporal primitives and whose geodesic metric predicts forecast reliability. These findings provide both a structural account of what large pretrained models learn from diverse time series corpora, and a practical zero-shot reliability signal that requires no retraining or label access.

Our approach complements attribution-based XAI for time series [Ismail et al., 2020, Theissler et al., 2022] by operating at the representation level rather than the prediction level. Where gradient-based attribution asks “which input timesteps matter for this forecast?”, LGF asks “what structure has the model learned to organise its embedding space?”—a mechanistic question that attribution methods cannot answer. The geodesic reliability protocol further bridges geometry and practice, providing a deployment criterion that is architecture-agnostic and label-free.

Limitations. Our analysis covers two TSFM architectures and four datasets; findings may not generalise to state-space-based TSFMs or domains with extreme non-stationarity. The TSFM vs. baseline comparison is structurally unmatched in pretraining-data volume, and while the data-scale ablation provides a partial control, we cannot fully disentangle training-data scale from architectural effects. The geodesic-forecast correlation is observational; causal claims require interventional experiments beyond the scope of this work. Future work should extend LGF to decoder-layer analysis for autoregressive TSFMs and to cross-domain transfer experiments that directly test whether geometric regularity predicts transferability.

Ethics Declaration

This work analyses publicly available pretrained model checkpoints and benchmark datasets. No human subjects, personal data, or proprietary data were involved. No known dual-use risks apply.

Data Availability

Electricity (UCI ML Repository), ETTh1 (ETDataset), Weather (Monash Forecasting Repository), and METR-LA (DCRNN repository) are publicly available. Pretrained weights are accessed via Hugging Face Hub. Code will be released at [anonymous repository].

Author Contributions (CRediT)

Conflict of Interest

The authors declare no competing interests.

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